

Service Parts Reference

Standard parts breakdown for all 23 canonical services — what goes on the invoice for each job. Vehicle-agnostic: the parts a service consumes are consistent across cars; only brand and spec change.

INTERNAL HANDOFF · BACKEND / DATA / SHOP ONBOARDING · MAY 2026

- **Core** — on essentially every invoice
- **As-needed** — situational (the discovery items that make a *range*)
- **Labor-only** — no parts billed

Labor-only services — no parts on the invoice

1 · Diagnostic Scan Scan-tool / computer time only.	2 · Pre-Purchase Inspection Inspection only. Excluded from MVP.
3 · Check Engine Light Diagnosis Diagnosis only; the fix is a separate service.	4 · State Inspection (NY) Inspection only. Sticker is a regulated fee, not a part.
5 · Emissions Test (NY) Test only.	12 · Tire Rotation Repositioning only (weights only if re-balanced).
14 · Wheel Alignment Adjustment only. As-needed: cam/alignment-bolt kit if out of range.	19 · Battery Test Load / alternator test only.

Parts-bearing services

6 Oil Change

CORE

- Engine oil (qty per spec)
- Oil filter

CONDITIONAL

- Drain-plug crush washer (where equipped — single-use)
- Oil-filter housing cap O-ring (cartridge-filter engines)

8 Spark Plugs

CORE

- Spark plugs (one per cylinder)

AS-NEEDED

- Ignition coils (if worn)
- Intake manifold / plenum gasket (engines needing manifold removal — many V6/V8 & DI)

10 Coolant Flush

CORE

- Coolant / antifreeze (concentrate + distilled water, or pre-mix)

AS-NEEDED

- Cooling-system flush chemical
- Thermostat + gasket (only if found bad)

13 Tire Balance

CORE

- Wheel weights (only consumable; rest is labor)

7 Filter Replacement

air + cabin

CORE

- Engine air filter
- Cabin air filter

9 Timing Belt

belt-driven only · N/A on chain

CORE

- Timing belt
- Belt tensioner
- Idler pulley(s)
- Camshaft & crankshaft seals

USUALLY BUNDLED (KIT)

- Water pump + coolant (if belt-driven)
- Accessory / serpentine belt

11 Transmission Service

two variants

DRAIN & FILL

- Transmission fluid (ATF / CVT per spec)
- Drain-plug crush washer

FULL PAN SERVICE ADDS

- Transmission filter (if serviceable)
- Pan gasket or RTV
- CVT: internal mesh screen + external filter w/ O-ring on some

15 Tire Replacement

separate tire flow

CORE

- Tire(s)
- TPMS service / valve kit per wheel (core, grommet, seal, nut, cap — standard on TPMS cars)
- Wheel weights (balancing)
- Tire disposal (fee)

16 Brake Pad Replacement

per axle

CORE

- Brake pads
- Caliper / brake grease

STANDARD / AS-NEEDED

- Hardware kit — clips, shims (often included with premium pads; replace if worn)
- Wear sensor (most BMW / Euro)

17 Rotor Replacement

per axle · pads together

CORE

- Brake rotors (2 per axle)
- Brake pads
- Caliper / brake grease

STANDARD / AS-NEEDED

- Hardware kit
- Wear sensor (Euro / BMW)

18 Brake Fluid Flush

CORE

- Brake fluid (DOT 3 / 4 / 4 LV per spec) — fluid only

20 Battery Replacement

CORE

- Battery

AS-NEEDED

- Anti-corrosion terminal washers / dielectric grease
- Terminal clamp (if corroded)
- BMW / Euro registration & coding = labor, not a part

21 Power Steering Flush

N/A on electric PS

CORE

- Power steering fluid (per spec)

AS-NEEDED

- Reservoir / in-line filter (some vehicles)
- Reservoir cap O-ring

22 Differential Service

CORE

- Gear oil (GL-5 hypoid per spec)
- Drain & fill plug crush washers

AS-NEEDED

- Friction modifier (LSD, if not pre-blended)
- Diff cover gasket / RTV (only if cover removed)

23 Fuel System / Induction Service

parts scale with depth of service

LIGHT — CHEMICAL

- Fuel-system / induction cleaner additive

MEDIUM — INDUCTION

- Intake / throttle cleaning spray

DEEP — WALNUT BLAST

- Walnut media (reusable — not consumed)
- Intake manifold gaskets (manifold removed)
- Throttle-body cleaner

Verification notes (two research sweeps)

Confirmed correct as the industry-standard breakdown. Four lines are standard-but-conditional rather than universal — they vary by vehicle design and shop practice, and align with the discovery items that drive a service to *range* instead of *lock*:

1. **Oil-change crush washer** — manufacturer-spec dependent; single-use where equipped (not every model uses one).
2. **Transmission service** — drain-and-fill (fluid + washer) vs. full pan service (adds filter + gasket); CVTs may add internal/external filters.
3. **Brake hardware** — clips/shims are best practice; often bundled with premium pads, and undamaged factory shims may be reused per the service manual.
4. **TPMS / valve kit on tire replacement** — standard best practice on TPMS-equipped cars ($\approx$ all US vehicles 2008+), but shop-policy dependent.